



LESSON SUMMARY

☞ Both concepts involve **line of sight** + **horizontal line**

☞ They form **right-angled triangles**

- Angle of elevation → looking **UP**
- Angle of depression → looking **DOWN**

☞ Most used formula:

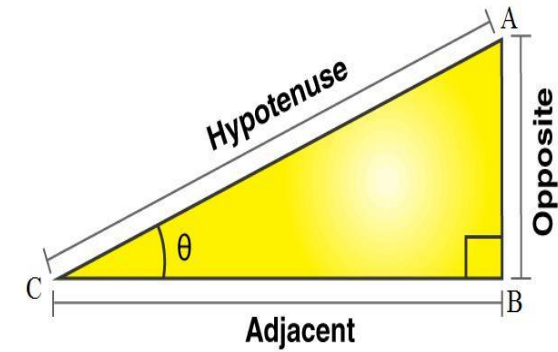
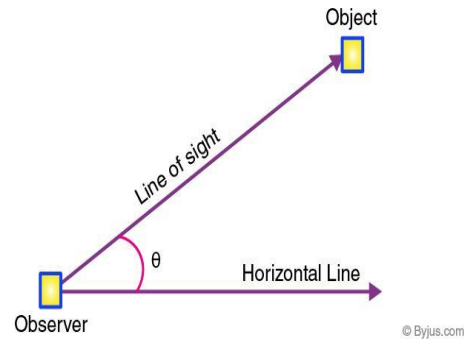
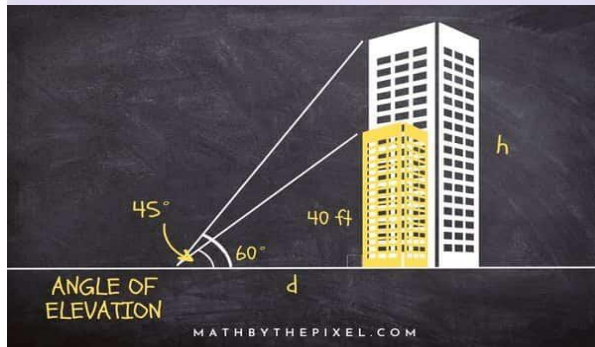
$$\tan \theta = \frac{\text{height}}{\text{distance}}$$

◆ A. ANGLE OF ELEVATION

📖 Definition

Angle between horizontal line and line of sight when looking **UP**.

📊 Visual Representation



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Key Features

- Object is **above eye level**
- Angle is **above horizontal**
- Forms a **right triangle**

Real-Life Examples

- Looking at a **building top**
- Viewing a **mountain peak**
- Watching a **plane**
- Flying a **kite**

◆ B. ANGLE OF DEPRESSION

📖 Definition

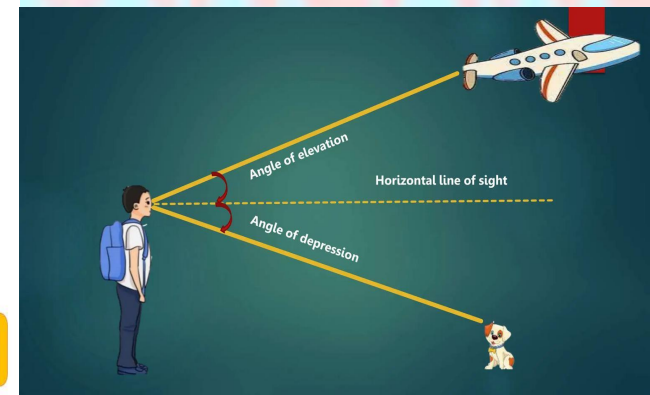
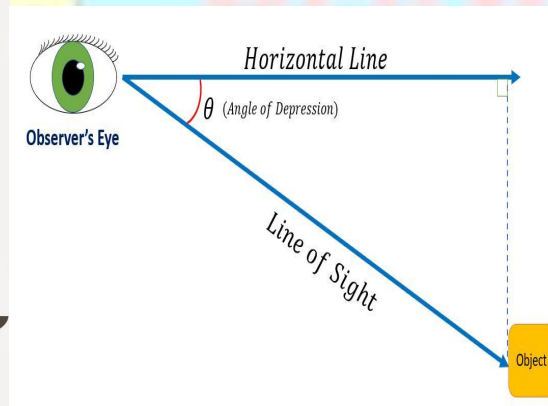
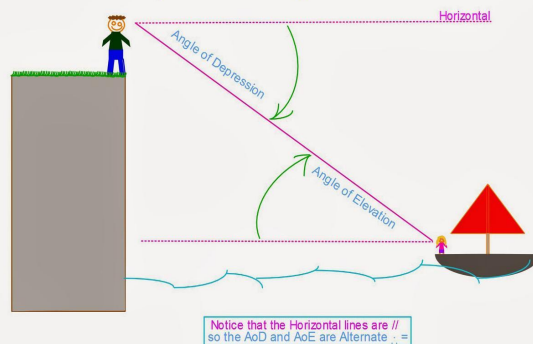
Angle between horizontal line and line of sight when looking **DOWN**.

🖼 Visual Representation

Elevation and Depression:

Elevation is the angle UP from the horizontal

Depression is the angle DOWN from the horizontal



🔑 Key Features



- Object is **below** eye level
- Angle is **below** horizontal
- Also forms a **right triangle**

◆ C. RELATIONSHIP (VERY IMPORTANT)

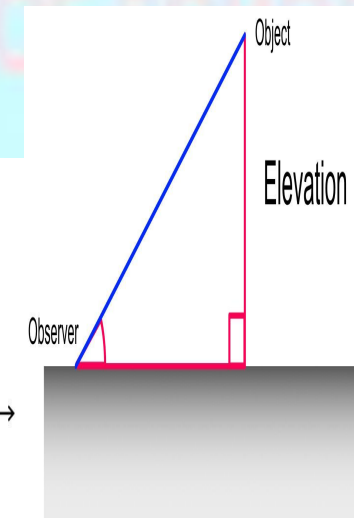
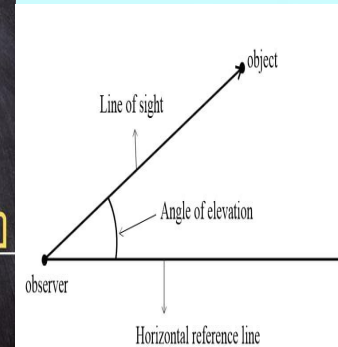
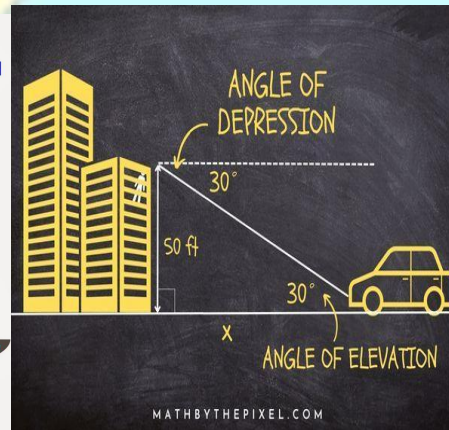
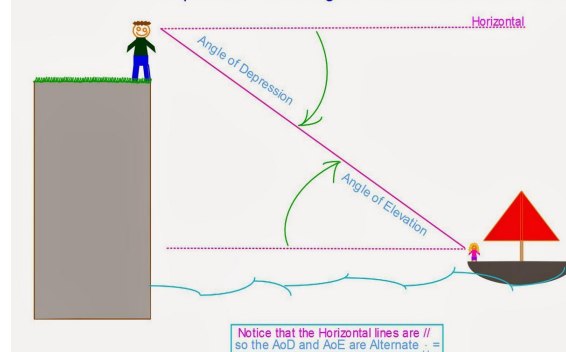
📖 Key Idea

👉 Angle of elevation = Angle of depression

🏠 Alternate Angles Concept

Elevation and Depression:

Elevation is the angle UP from the horizontal
Depression is the angle DOWN from the horizontal



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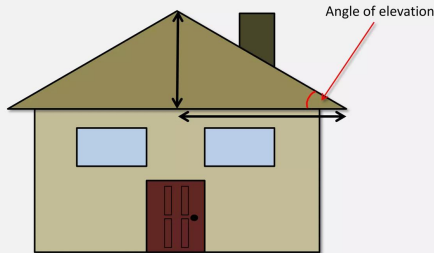
Why?

- Horizontal lines are **parallel**
- Line of sight is a **transversal**
- Alternate angles are equal

◆ D. CONSTRUCTION OF DIAGRAMS

⊗ Steps: Angle of Elevation

Angle of Elevation



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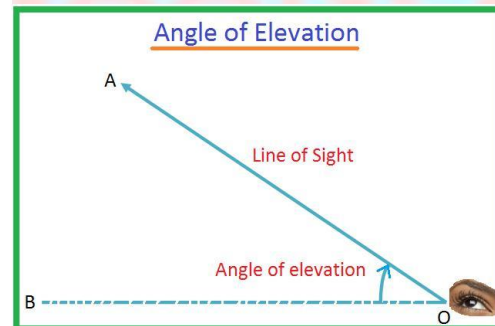
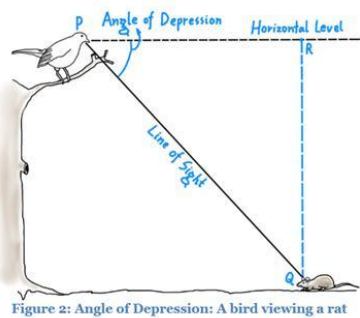
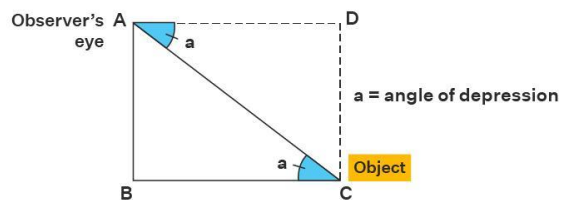
Steps:

1. Draw horizontal line
2. Mark observer

3. Draw vertical (height)
4. Join line of sight
5. Label θ , h, d

⊗ Steps: Angle of Depression

Angle of Depression in a Triangle



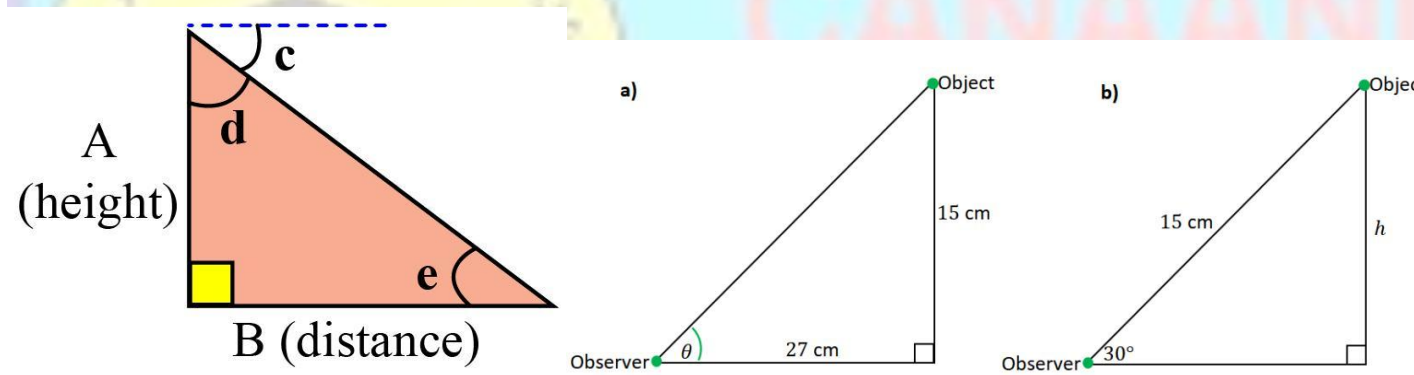
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Steps:

1. Draw horizontal at observer
2. Mark object below
3. Draw vertical
4. Draw line of sight
5. Label clearly

◆ E. TRIGONOMETRIC APPLICATION

▮ Triangle Setup



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To Find:

Formula

When to Use:

Height (h)

$$h = d \times \tan \theta$$

You know distance and angle

Distance (d)

$$d = h / \tan \theta$$

You know height and angle

To Find:

Formula

When to Use:

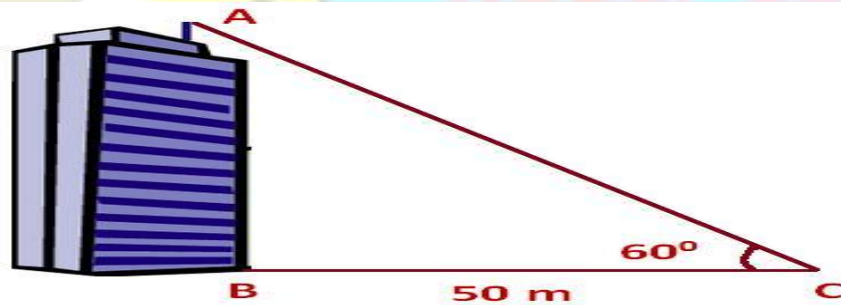
Angle (θ)

$$\theta = \tan^{-1} (h / d)$$

You know height and distance

WORKED EXAMPLES

✓ **Example 1: Height of Building:** From a point 50 m from the base of a building, the angle of elevation to the top is 60° . Find the height of the building.



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Given:

- Distance = 50 m



- Angle = 60°

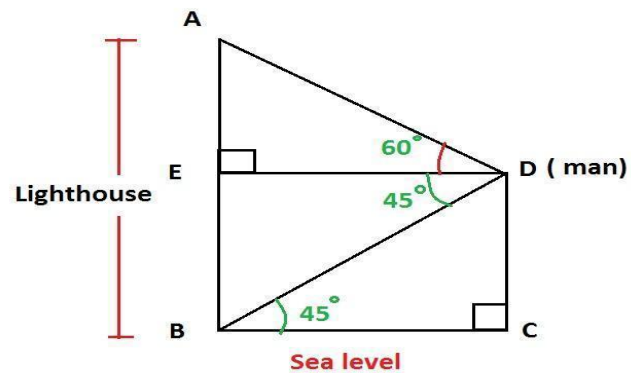
$$h = d \tan \theta$$

$$h = 50 \times \tan(60^\circ)$$

$$h = 50 \times 0.3200$$

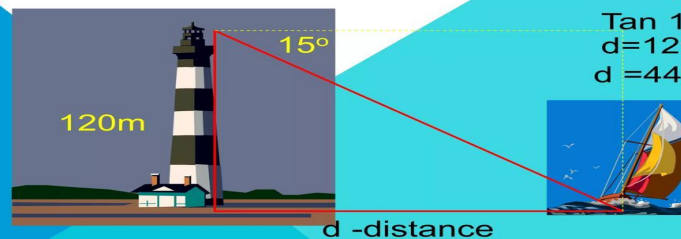
$$h = 16.00 \text{ m} \Rightarrow \text{Answer: } 16.00 \text{ m}$$

Example 2: From the top of a lighthouse 60 m high, the angle of depression to a boat is 25° . How far is the boat from the base of the lighthouse? Diagram



Real-Life Problem on right triangle

From the lighthouse 120m above the sea, the angle of depression of a boat is 15° . How far is the boat from the lighthouse?



$$15^\circ = 120/d$$

$$\tan 15^\circ = 120/d$$

$$d = 120 / \tan 15^\circ$$

$$d = 448 \text{ m}$$

$$d = h / \tan \theta$$

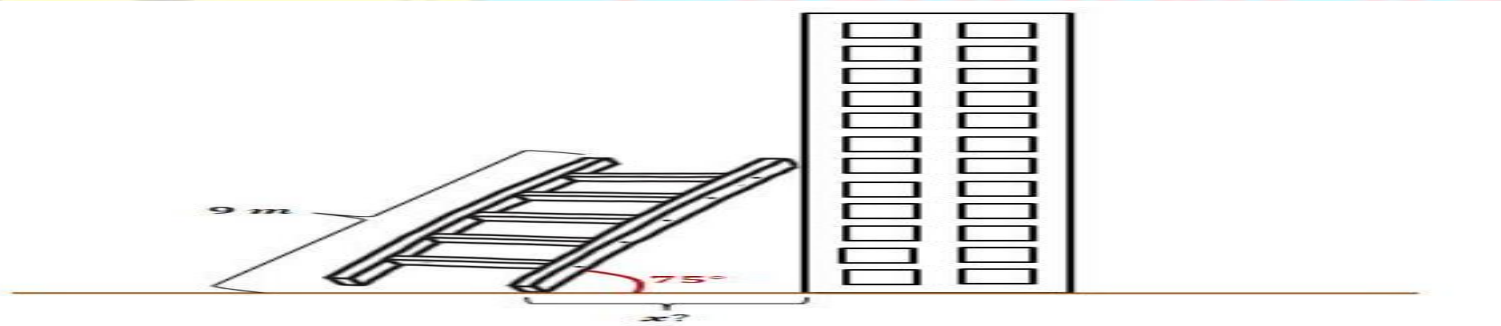


$$d = 60 / \tan(25^\circ)$$

$$d = 60 / 0.4663$$

$$d = 128.65 \text{ m} \Rightarrow \text{Answer: } 128.65 \text{ m}$$

Example 3: A ladder 10 m long leans against a wall. If the ladder makes an angle of 65° with the ground, how high up the wall does it reach?



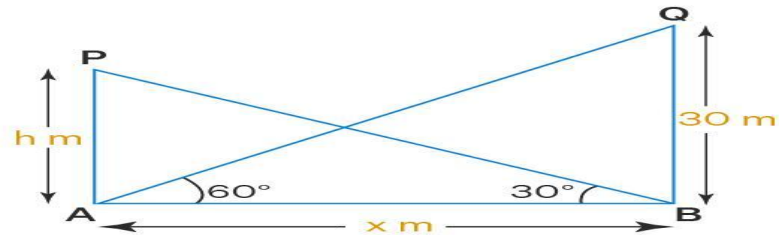
$$h = L \sin \theta$$

$$h = 10 \times \sin(65^\circ)$$

$$h = 10 \times 0.9063$$

$$h = 9.06 \text{ m}$$

Example 4: From a point on the ground, the angle of elevation to the top of a tower is 30° . From a point 40 m closer to the tower, the angle of elevation is 45° . Find the height of the tower.



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Solution: Let distance from B to tower = x

Then distance from A to tower = $x + 40$

From point B: $\tan 45^\circ = \frac{h}{x} \quad 1 = \frac{h}{x}$ (since $\tan 45^\circ = 1$) $h = x \dots (1)$

From point A: $\tan 30^\circ = \frac{h}{(x + 40)} \quad 0.5774 = \frac{h}{(x + 40)} \quad h = 0.5774(x + 40) \dots (2)$

Substitute (1) into (2): $x = 0.5774(x + 40) \quad x = 0.5774x + 23.096 \quad x - 0.5774x = 23.096 \quad 0.4226x = 23.096 \quad x = 54.64 \text{ m}$

Therefore: $h = x = 54.64 \text{ m}$

Example 5: A man standing 80 m from a tower observes the angle of elevation to the top to be 38° . Find: (a) The height of the tower (b) The distance from the man to the top of the tower (line of sight)

Given:

Adjacent side (horizontal distance) = 80 m

Angle (θ) = 38°

Solution: (a) $\tan 38^\circ = \frac{h}{80} \quad h = 80 \times \tan 38^\circ \quad h = 80 \times 0.7813 \quad h = 62.50 \text{ m}$

(b) $\cos 38^\circ = \frac{80}{s} \quad s = \frac{80}{\cos 38^\circ} \quad s = \frac{80}{0.7880} \quad s = 101.52 \text{ m}$

Answer: (a) Height = 62.50 m (b) Line of sight = 101.52m

PART 1: FINDING HEIGHT (h)



Formula:

text

$$h = d \times \tan \theta$$

Where:

h = vertical height (opposite side)

d = horizontal distance (adjacent side)

θ = angle

Substitution:

Example 5: A man standing 80 m from a tower observes the angle of elevation to the top to be 38° . Find: (a) The height of the tower (b) The distance from the man to the top of the tower (line of sight) $h = 80 \times \tan 38^\circ$

Calculation:

$$\tan 38^\circ = 0.7813$$

$$h = 80 \times 0.7813$$

$$h = 62.50 \text{ m}$$

Answer for Height:

The height is 62.50 metres.



PART 2: FINDING SLANT DISTANCE (s) Formula:

$$s = d / \cos \theta$$

Where:

s = slant distance (hypotenuse)

d = horizontal distance (adjacent side)

θ = angle

Substitution:

$$s = 80 / \cos 38^\circ$$

Calculation:

$$\cos 38^\circ = 0.7880$$

$$s = 80 / 0.7880$$

$$s = 101.52 \text{ m}$$

Answer for Slant Distance:

The slant distance is 101.52 metres.



✓ **Example 6: Two Cars Problem** Example 6: From the top of a building 40 m high, the angle of depression to a car is 28° , and the angle of depression to another car beyond the first is 18° . Find the distance between the two cars.

Distance between cars = **47.87 m**

✓ **Example 7: Plane Speed**

Speed = **472.44 km/h**

✓ **Example 8: Two Buildings**

Height of tall building = **41.01 m**

CLASS ACTIVITIES

1. Build a **clinometer**
2. Draw:
 - Elevation diagram
 - Depression diagram
3. Solve:
 - Angle = 28° , distance = 100 m

? EVALUATION QUESTIONS



CANAANLAND
MODEL GROUP OF SCHOOLS

GOD IS OUR STRENGTH

1. Define angle of elevation
2. Explain elevation = depression
3. Solve height problems
4. Draw labeled diagram
5. Solve real-life scenario

ASSIGNMENT

Includes:

- Height calculations
- Distance problems
- Ladder problems
- Aircraft problems

FINAL SUMMARY

- ✓ Angle of elevation → look UP
- ✓ Angle of depression → look DOWN
- ✓ They are **equal (alternate angles)**
- ✓ Use:



DERIVED FORMULAS

From the basic formula $\tan \theta = h / d$, we can derive:

To Find:	Formula
Height (h)	$h = d \times \tan \theta$
Distance (d)	$d = h / \tan \theta$
Angle (θ)	$\theta = \tan^{-1} (h / d)$

EXAMPLE USE

Given:

-

Distance (d) = 50 m

-



-

Angle (θ) = 32°

-

Calculation:

text

$$\tan 32^\circ = h / 50h = 50 \times \tan 32^\circ h = 50 \times 0.6249h = 31.25 \text{ m}$$

✓ Applications:

- Buildings
- Aviation
- Navigation
- Surveying